TOPIC:

BANK CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Internship Project Report

Submitted in partial fulfillment of the requirements for the award of the Certificate of

Internship of Business Analytics

Submitted By

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Internship Organization

Unified Mentor Pvt. Ltd.

Business Analytics Internship

Submitted To

Unified Mentor pvt. Ltd.

CERTIFICATE

This is to certify that the project report entitled **"Bank Customer Churn Prediction using Machine Learning"** has been successfully completed by **Mr. Ashutosh Sharma** as part of the **Business Analytics Internship at Unified Mentor**.

The project demonstrates the application of Business Analytics, Machine Learning, Data Visualization, and Streamlit Dashboard Development in predicting customer churn within the banking sector. The work carried out reflects the candidate's understanding of data preprocessing, exploratory data analysis, feature engineering, predictive modeling, explainable artificial intelligence, and deployment of an interactive analytics dashboard.

To the best of our knowledge, this work is original and has been completed under the internship program for academic and learning purposes.

Date: ____________

Place: ___________

Internship Organization

Unified Mentor

DECLARATION

I, **Ashutosh Sharma**, hereby declare that the project report entitled **"Bank Customer Churn Prediction using Machine Learning"** is an original piece of work carried out by me during my **Business Analytics Internship at Unified Mentor**.

This report has been prepared solely for academic purposes and has not been submitted, either wholly or partly, for the award of any other degree, diploma, or certification at any university or institution.

I further declare that all sources of information used in this project have been properly acknowledged and referenced.

Ashutosh Sharma

Date: ____________

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Unified Mentor** for providing me with the opportunity to undertake the **Business Analytics Internship**, which enabled me to gain valuable practical exposure to Machine Learning, Business Analytics, and Dashboard Development.

I extend my heartfelt thanks to the mentors and support team at Unified Mentor for their guidance, encouragement, and valuable feedback throughout the internship. Their continuous support helped me successfully complete this project and strengthen my practical understanding of data analytics concepts.

I am also grateful to **Amity University Online** for providing a strong academic foundation in Business Analytics, which greatly contributed to the successful completion of this project.

Finally, I would like to thank my family and well-wishers for their constant encouragement, motivation, and support throughout my academic journey and during the completion of this internship project.

Ashutosh Sharma

ABSTRACT

Customer churn prediction has become a critical application of data analytics and machine learning in the banking industry. Retaining existing customers is significantly more cost-effective than acquiring new ones, making churn prediction an essential business strategy. This project, **"Bank Customer Churn Prediction using Machine Learning,"** was successfully completed as part of the **Data Analytics Internship at Unified Mentor**. The primary objective of the project was to develop an intelligent predictive system capable of identifying customers who are likely to discontinue banking services based on historical customer information.

The project followed a systematic data analytics workflow that included data collection, preprocessing, exploratory data analysis (EDA), feature engineering, predictive modeling, model evaluation, and dashboard development. The dataset consisted of demographic, financial, and customer engagement attributes such as credit score, age, tenure, account balance, geographical location, number of products, active membership status, and estimated salary. After cleaning the dataset and preparing it for analysis, several machine learning techniques were evaluated to identify the most suitable predictive model.

To improve model transparency and interpretability, Feature Importance Analysis and SHAP (SHapley Additive Explanations) were implemented to explain how individual features influence customer churn predictions. These explainability techniques help business users understand the reasoning behind model decisions and support informed strategic planning.

An interactive dashboard was developed using **Streamlit**, enabling users to predict customer churn in real time by entering customer information through a user-friendly interface. The dashboard also includes key performance indicators (KPIs), probability distribution visualizations, feature importance analysis, SHAP explainability plots, and a What-if Scenario Simulator that allows users to modify customer attributes and instantly observe changes in churn probability.

The final application was version-controlled using GitHub and deployed on Streamlit Cloud, making it accessible through a web browser. This project demonstrates practical implementation of Business Analytics, Machine Learning, Data Visualization, Explainable AI, and Interactive Dashboard Development using Python technologies such as Pandas, NumPy, Scikit-learn, Plotly, SHAP, and Streamlit.

Overall, the developed system provides an effective decision-support solution for banking organizations by enabling early identification of high-risk customers, supporting targeted retention campaigns, and enhancing data-driven decision-making.

Keywords: Customer Churn Prediction, Machine Learning, Business Analytics, Streamlit, Data Visualization, Feature Engineering, SHAP, Explainable AI, Banking Analytics.

Executive Summary

Customer retention has become one of the most significant challenges in the banking industry, where acquiring new customers is considerably more expensive than retaining existing ones. Customer churn not only reduces revenue but also affects long-term business growth and customer satisfaction. Therefore, accurately identifying customers who are likely to leave enables banks to implement proactive retention strategies and improve overall business performance.

This project, titled **"Bank Customer Churn Prediction using Machine Learning"**, was successfully completed as part of the **Data Analytics Internship at Unified Mentor**. The project focuses on developing an end-to-end predictive analytics solution capable of estimating the likelihood of customer churn based on demographic, financial, and behavioral information.

The project followed the complete data analytics lifecycle, beginning with data collection and preprocessing. The dataset was cleaned, missing values were handled, and categorical variables were encoded to prepare the data for analysis. Exploratory Data Analysis (EDA) was performed to identify significant patterns, customer behaviors, and relationships among various features influencing churn. Additional feature engineering techniques were applied to improve predictive performance by creating meaningful derived variables.

Several machine learning algorithms were evaluated to identify the most suitable model for customer churn prediction. The selected model was trained, validated, and optimized using appropriate evaluation metrics to ensure reliable prediction performance. Model interpretability techniques, including **Feature Importance Analysis** and **SHAP (SHapley Additive Explanations)**, were incorporated to explain how individual features contribute to churn predictions, thereby improving transparency and business understanding.

To make the predictive model accessible to business users, an interactive web application was developed using **Streamlit**. The dashboard provides real-time customer churn prediction, key performance indicators, probability distribution visualizations, feature importance analysis, SHAP explainability plots, and a What-if Scenario Simulator that enables users to modify customer attributes and observe changes in churn probability. This interactive environment supports informed decision-making for customer retention strategies.

The complete solution was version-controlled using GitHub and successfully deployed on Streamlit Cloud, making the application publicly accessible for demonstration and portfolio purposes. The project demonstrates practical implementation of Machine Learning, Data Analytics, Data Visualization, Explainable AI, and Dashboard Development while integrating multiple Python libraries including Pandas, NumPy, Scikit-learn, Plotly, SHAP, and Streamlit.

Overall, the project provides an effective decision-support system that assists banking institutions in identifying high-risk customers, understanding key churn drivers, and designing targeted customer retention initiatives. It also showcases the practical application of Business

Analytics concepts in solving real-world business problems and highlights the value of predictive analytics in modern banking.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The banking industry has experienced rapid digital transformation over the past decade, leading to increased competition among financial institutions. Customers today have numerous banking options and can easily switch to competing organizations if their expectations regarding service quality, pricing, digital experience, or customer support are not met. As a result, customer retention has become one of the highest priorities for banks seeking long-term profitability and sustainable growth.

Customer churn refers to the phenomenon in which customers discontinue using a company's products or services. In the banking sector, customer churn directly impacts revenue generation, operational costs, and customer lifetime value. Studies consistently show that acquiring a new customer is considerably more expensive than retaining an existing one. Therefore, identifying customers who are at high risk of leaving enables banks to implement timely retention strategies and improve overall customer satisfaction.

Traditional approaches to customer retention primarily relied on manual analysis and historical observations, which are often time-consuming and less accurate. However, advances in Machine Learning and Business Analytics have enabled organizations to develop predictive models capable of analyzing large volumes of customer data and accurately estimating churn probability. These predictive systems assist decision-makers by identifying hidden patterns and relationships that may not be apparent through conventional analysis.

This project focuses on developing an end-to-end **Bank Customer Churn Prediction System** using Machine Learning techniques. The solution integrates data preprocessing, exploratory data analysis, feature engineering, predictive modeling, explainable artificial intelligence, and interactive dashboard development into a unified analytics platform. The application predicts customer churn probability based on multiple demographic, financial, and behavioral characteristics and presents the results through a user-friendly Streamlit dashboard.

The developed dashboard not only predicts customer churn but also provides meaningful visualizations, feature importance analysis, SHAP explainability plots, and a What-if Scenario Simulator that allows users to evaluate how changes in customer characteristics influence churn probability. Such capabilities help banking professionals make informed business decisions and design targeted customer retention initiatives.

This project demonstrates the practical application of Business Analytics concepts and Machine Learning methodologies in solving a real-world business problem while emphasizing the importance of explainable and interactive analytics for effective decision support.

CHAPTER 2: PROBLEM STATEMENT

2.1 Problem Statement

Customer churn is one of the major challenges faced by banks and financial institutions. When customers discontinue their relationship with a bank, it leads to reduced revenue, increased customer acquisition costs, and lower profitability. In today's competitive banking environment, customers can easily switch to competitors due to better financial products, improved digital services, attractive offers, or enhanced customer support.

Traditional methods of identifying customers who are likely to leave often rely on manual analysis and historical reports. These approaches are time-consuming, less accurate, and incapable of detecting complex relationships among multiple customer attributes. Consequently, banks require an intelligent and automated system capable of predicting customer churn before customers decide to leave.

The primary challenge addressed in this project is the development of a Machine Learning-based predictive system that can accurately identify customers at risk of churning. The solution must also provide meaningful insights into the factors influencing churn while enabling business users to interact with the prediction results through a user-friendly dashboard.

To address this challenge, the project integrates data preprocessing, feature engineering, machine learning, explainable AI, and interactive visualization within a Streamlit dashboard. This enables banks to proactively identify high-risk customers, design targeted retention strategies, and support data-driven decision-making.

CHAPTER 3: PROJECT OBJECTIVES

3.1 Primary Objective

The primary objective of this project is to design and develop an end-to-end **Bank Customer Churn Prediction System** using Machine Learning and Business Analytics techniques that accurately predicts whether a customer is likely to leave the bank and presents the results through an interactive Streamlit dashboard.

3.2 Specific Objectives

The specific objectives of the project are:

- To collect and preprocess customer data for predictive analysis.
- To perform Exploratory Data Analysis (EDA) to identify important customer behavior patterns.
- To engineer meaningful features that improve model performance.
- To develop a Machine Learning model capable of predicting customer churn accurately.
- To evaluate model performance using appropriate classification metrics.
- To analyze model predictions using Feature Importance and SHAP Explainability techniques.
- To build an interactive Streamlit dashboard for real-time customer churn prediction.
- To implement a What-if Scenario Simulator for analyzing changes in churn probability.
- To deploy the application on Streamlit Cloud for public accessibility.
- To support banking organizations in designing proactive customer retention strategies through data-driven insights.

CHAPTER 4: PROJECT SCOPE

4.1 Scope of the Project

The scope of this project is focused on developing a practical Machine Learning application for predicting customer churn in the banking sector. The project demonstrates the complete analytics lifecycle, from data preprocessing to deployment, using real-world customer data.

The project includes:

- Customer data preprocessing and cleaning.
- Exploratory Data Analysis (EDA).
- Feature engineering and transformation.
- Machine Learning model development and evaluation.
- Explainable AI using Feature Importance and SHAP analysis.
- Interactive Streamlit dashboard development.
- Customer churn probability prediction.
- What-if Scenario Simulator.
- Deployment using GitHub and Streamlit Cloud.

Although the developed system effectively predicts customer churn, it is designed for analytical and educational purposes. Future improvements may include integration with live banking databases, real-time prediction APIs, automated customer segmentation, personalized recommendation systems, and cloud-based enterprise deployment.

CHAPTER 5: LITERATURE REVIEW

5.1 Introduction

Customer churn prediction has become one of the most important applications of Business Analytics and Machine Learning in the banking industry. As competition among financial institutions continues to increase, retaining existing customers has become a strategic priority because customer acquisition is significantly more expensive than customer retention. Organizations increasingly use predictive analytics to identify customers who are likely to discontinue their services and implement proactive retention strategies.

Machine Learning has transformed churn prediction by enabling organizations to analyze large volumes of customer data and identify hidden patterns that traditional statistical techniques often fail to detect. Recent advancements in Explainable Artificial Intelligence (XAI) have further improved the transparency and interpretability of predictive models, making them more reliable for business decision-making.

5.2 Customer Churn Prediction in Banking

Customer churn prediction aims to estimate the probability that a customer will discontinue using banking services. Researchers have shown that demographic characteristics, financial behavior, transaction history, customer engagement, and product usage significantly influence customer retention.

Several studies have demonstrated that machine learning models outperform traditional statistical approaches because they can capture complex relationships among multiple variables. Predictive analytics enables financial institutions to identify high-risk customers before they leave, allowing businesses to implement personalized retention campaigns, loyalty programs, and targeted marketing strategies.

5.3 Machine Learning Techniques for Churn Prediction

Numerous machine learning algorithms have been applied to customer churn prediction, including Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), Gradient Boosting, and XGBoost.

Among these techniques, ensemble learning methods such as Random Forest and Gradient Boosting have demonstrated strong predictive performance due to their ability to reduce overfitting and improve classification accuracy. These models analyze multiple decision trees collectively to generate more reliable predictions.

Model performance is generally evaluated using classification metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These evaluation measures provide comprehensive insight into a model's predictive capability and its effectiveness in identifying churned customers.

5.4 Importance of Explainable Artificial Intelligence (XAI)

Although advanced machine learning models often achieve high prediction accuracy, they are frequently criticized for their limited interpretability. Explainable Artificial Intelligence (XAI) addresses this limitation by providing understandable explanations of model predictions.

Feature Importance Analysis identifies the variables that contribute most significantly to customer churn. SHAP (SHapley Additive Explanations) further enhances model interpretability by measuring the contribution of each feature to an individual prediction.

The integration of XAI improves transparency, builds stakeholder trust, and enables business managers to understand why specific customers are classified as high-risk.

5.5 Interactive Dashboard Development

Modern Business Analytics emphasizes the importance of presenting predictive insights through interactive dashboards rather than static reports. Dashboards improve decision-making by allowing users to explore data visually and interact with predictive models.

Streamlit has become one of the most widely used frameworks for developing data science applications because it enables rapid deployment of interactive machine learning dashboards with minimal coding effort. Streamlit applications support real-time prediction, data visualization, feature exploration, and scenario analysis, making them valuable decision-support tools.

5.6 Research Gap

Although existing research has extensively explored customer churn prediction using machine learning, many studies primarily focus on model development and evaluation while providing limited support for real-time business decision-making.

Few solutions integrate predictive analytics, explainable AI, interactive visualization, and deployment within a single application. This project addresses that gap by combining machine learning, feature importance analysis, SHAP explainability, interactive Streamlit dashboards, and What-if Scenario Simulation into one unified decision-support system.

5.7 Summary

The literature indicates that Machine Learning provides an effective solution for customer churn prediction, while Explainable AI improves model transparency and business understanding. Interactive dashboards further enhance the practical usability of predictive models by enabling decision-makers to explore insights and perform real-time analysis. Building upon these concepts, the present project develops a comprehensive customer churn prediction system that integrates predictive modeling, explainability, visualization, and deployment into a single end-to-end analytics solution.

CHAPTER 6: DATASET DESCRIPTION

6.1 Introduction

A high-quality dataset is the foundation of any successful machine learning project. The performance of predictive models depends on the quality, completeness, and relevance of the available data. For this project, a structured banking customer dataset containing demographic, financial, and behavioral information was used to predict customer churn. The dataset provides historical information about customers and whether they continued using the bank's services or discontinued their relationship.

The primary objective of using this dataset was to train a machine learning model capable of identifying customers with a high probability of churning. The dataset includes multiple features that influence customer behavior and support effective predictive analysis.

6.2 Dataset Overview

The project uses the **European Bank Customer Dataset**, which contains customer-level information collected from a banking institution. Each record represents an individual customer along with several demographic, financial, and account-related attributes.

The dataset contains information such as customer credit score, geographical location, gender, age, account tenure, account balance, number of banking products, credit card ownership, active membership status, estimated salary, and customer churn status.

The target variable used for prediction is **Exited**, where:

- **0** indicates that the customer remained with the bank.
- **1** indicates that the customer left the bank (churned).

6.3 Dataset Features

The dataset consists of the following important attributes:

Feature	Description
CreditScore	Customer's credit score
Geography	Country of the customer
Gender	Male or Female
Age	Customer's age
Tenure	Number of years with the bank

Balance	Current account balance
NumOfProducts	Number of banking products used
HasCrCard	Whether the customer owns a credit card
IsActiveMember	Whether the customer is an active member
EstimatedSalary	Annual estimated salary
Exited	Target variable indicating customer churn

6.4 Dataset Characteristics

The dataset contains both **categorical** and **numerical** variables, making it suitable for supervised classification.

Numerical Variables

- Credit Score
- Age
- Tenure
- Balance
- Number of Products
- Estimated Salary

Categorical Variables

- Geography
- Gender
- Credit Card Status
- Active Membership

The combination of these variables provides valuable insights into customer behavior and supports accurate churn prediction.

6.5 Target Variable

The target variable **Exited** is a binary classification variable.

Value	Meaning
0	Customer Stayed
1	Customer Churned

The machine learning model was trained to predict this variable based on customer characteristics.

6.6 Importance of the Dataset

The selected dataset provides comprehensive information that allows the model to identify patterns associated with customer churn. Features such as customer age, account balance, product usage, credit score, and engagement level significantly influence churn behavior.

Using this dataset enables financial institutions to understand customer risk profiles and implement proactive retention strategies before customers discontinue banking services.

6.7 Summary

The European Bank Customer Dataset forms the foundation of this project by providing relevant demographic, financial, and behavioral information required for churn prediction. Its combination of numerical and categorical features supports comprehensive exploratory analysis, feature engineering, predictive modeling, and business insight generation.

CHAPTER 7: DATA PREPROCESSING

7.1 Introduction

Raw data often contains inconsistencies, redundant information, and formats that are unsuitable for machine learning algorithms. Therefore, data preprocessing is an essential step in transforming raw data into a structured format suitable for analysis and predictive modeling.

In this project, several preprocessing techniques were applied to improve data quality and prepare the dataset for machine learning. These included handling missing values, removing irrelevant attributes, encoding categorical variables, and preparing the dataset for feature engineering and model training.

7.2 Data Cleaning

The initial stage involved examining the dataset for missing values, duplicate records, and inconsistent data entries.

The following preprocessing tasks were performed:

- Verified dataset completeness.
- Checked for missing values.
- Removed unnecessary columns.
- Validated data types.
- Ensured consistency across all attributes.

The dataset was found to be well-structured and required minimal cleaning before analysis.

7.3 Handling Missing Values

The dataset was inspected for null values across all columns. Missing values can negatively impact model performance by introducing bias and reducing predictive accuracy.

During preprocessing:

- Each feature was checked for null values.

- Appropriate validation confirmed that the dataset contained no significant missing information requiring imputation.

This ensured that the dataset maintained its integrity throughout the modeling process.

7.4 Encoding Categorical Variables

Machine learning algorithms require numerical input. Therefore, categorical attributes were transformed into numerical representations.

The following variables were encoded:

Variable	Encoding Method
Gender	Binary Encoding
Geography	One-Hot Encoding

Encoding categorical variables enabled the machine learning model to process customer demographic information effectively.

7.5 Feature Scaling

Feature scaling was applied where appropriate to ensure that numerical variables operated within comparable ranges and prevented features with larger values from dominating the learning process.

Scaling contributed to improved model stability and training performance.

7.6 Data Splitting

The processed dataset was divided into training and testing datasets.

The training dataset was used to develop the predictive model, while the testing dataset evaluated model performance on unseen data.

This approach ensured objective assessment of the model's predictive capability.

7.7 Summary

Data preprocessing transformed the raw banking dataset into a structured format suitable for machine learning. Cleaning, encoding, feature preparation, and dataset partitioning ensured high-quality input data for predictive modeling and improved the reliability of subsequent analyses.

CHAPTER 8: EXPLORATORY DATA ANALYSIS (EDA)

8.1 Introduction

Exploratory Data Analysis (EDA) is an essential phase of the data analytics process that helps in understanding the structure, characteristics, and relationships within the dataset before building predictive models. It enables analysts to identify patterns, detect anomalies, and gain meaningful insights that guide feature engineering and model selection.

In this project, EDA was performed using **Pandas, Matplotlib, and Plotly** to analyze customer demographics, financial attributes, and churn behavior. The visualizations generated during this phase helped identify the key factors influencing customer churn and provided valuable business insights.

8.2 Customer Churn Distribution

The first step was to analyze the overall distribution of customers who stayed with the bank versus those who churned.

A donut chart was created to visualize the proportion of churned and retained customers. The analysis revealed that most customers continued using the bank's services, while a smaller proportion exited. This indicates that the dataset is moderately imbalanced, which is common in churn prediction problems.

Business Insight:
The bank retains the majority of its customers, but identifying the minority group at risk of churn is essential for implementing proactive retention strategies.

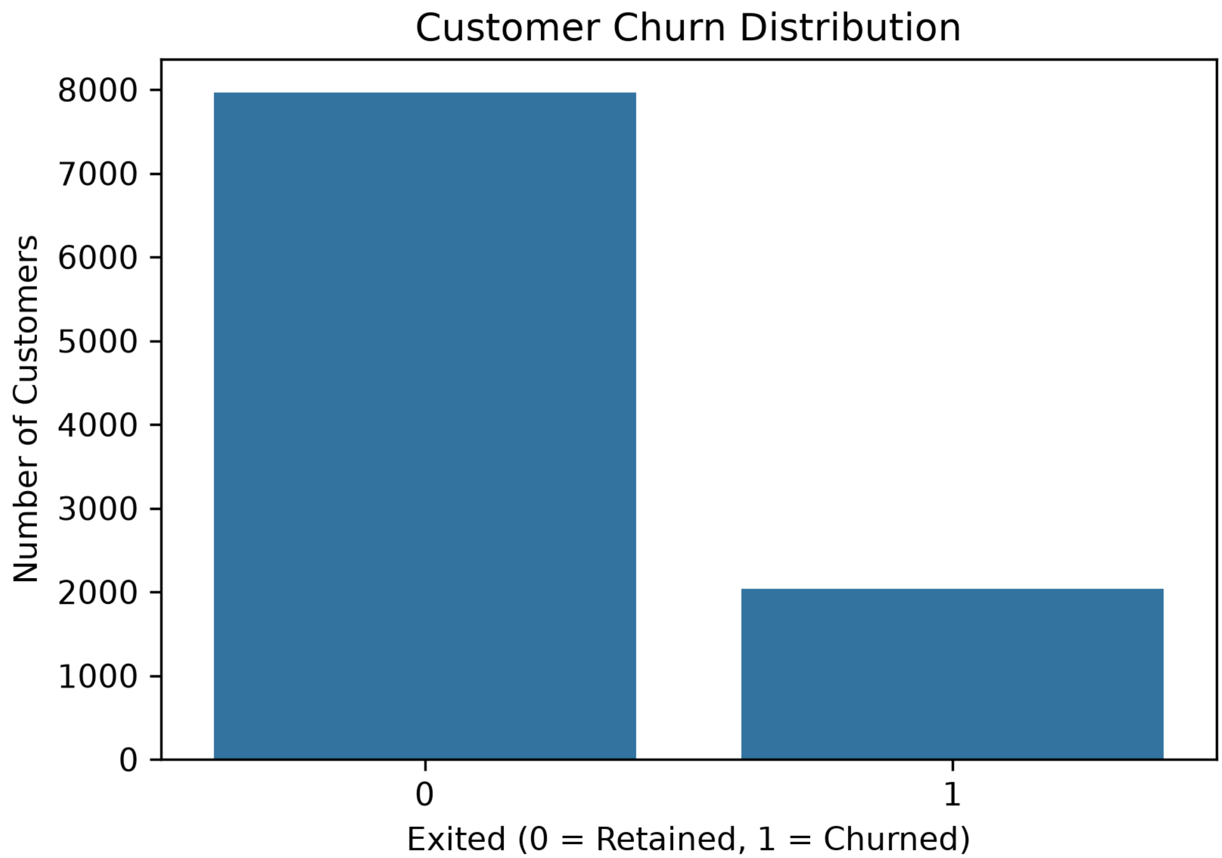

Figure 8.1: Customer Churn Distribution

8.3 Churn Rate by Geography

Customer churn was further analyzed across different geographical regions.

A bar chart was created to compare churn rates among France, Germany, and Spain. The analysis showed that churn behavior varies significantly across regions, indicating that geographical location influences customer retention.

Business Insight:
Regional differences in customer behavior suggest that banks should implement location-specific retention strategies rather than adopting a uniform approach.

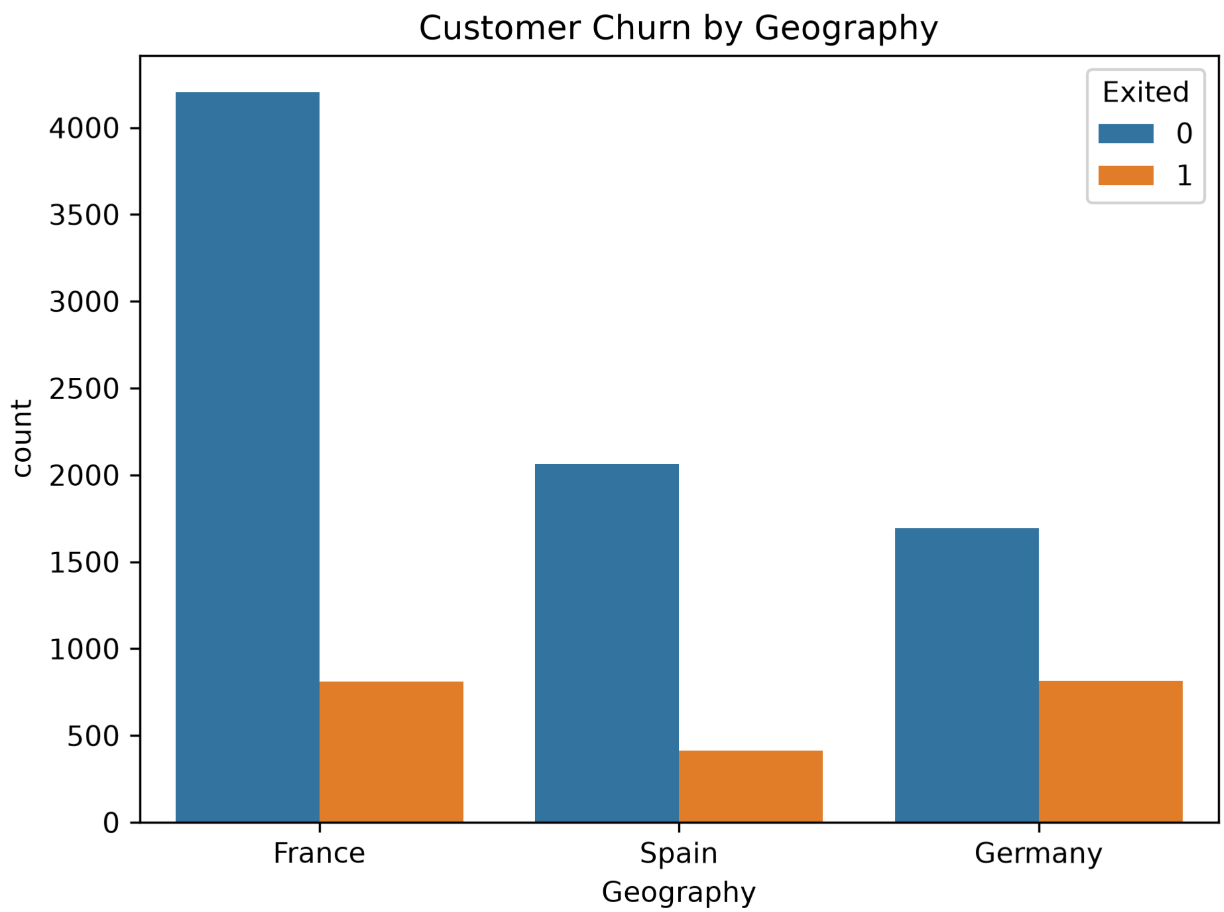

Figure 8.2: Churn Rate by Geography

8.4 Age Distribution Analysis

Customer age was examined using a histogram segmented by churn status.

The analysis demonstrated that churn probability tends to increase among older customer groups, while younger customers generally exhibit higher retention rates.

Business Insight:
Age is an important predictor of customer churn and should be considered when designing personalized customer engagement strategies.

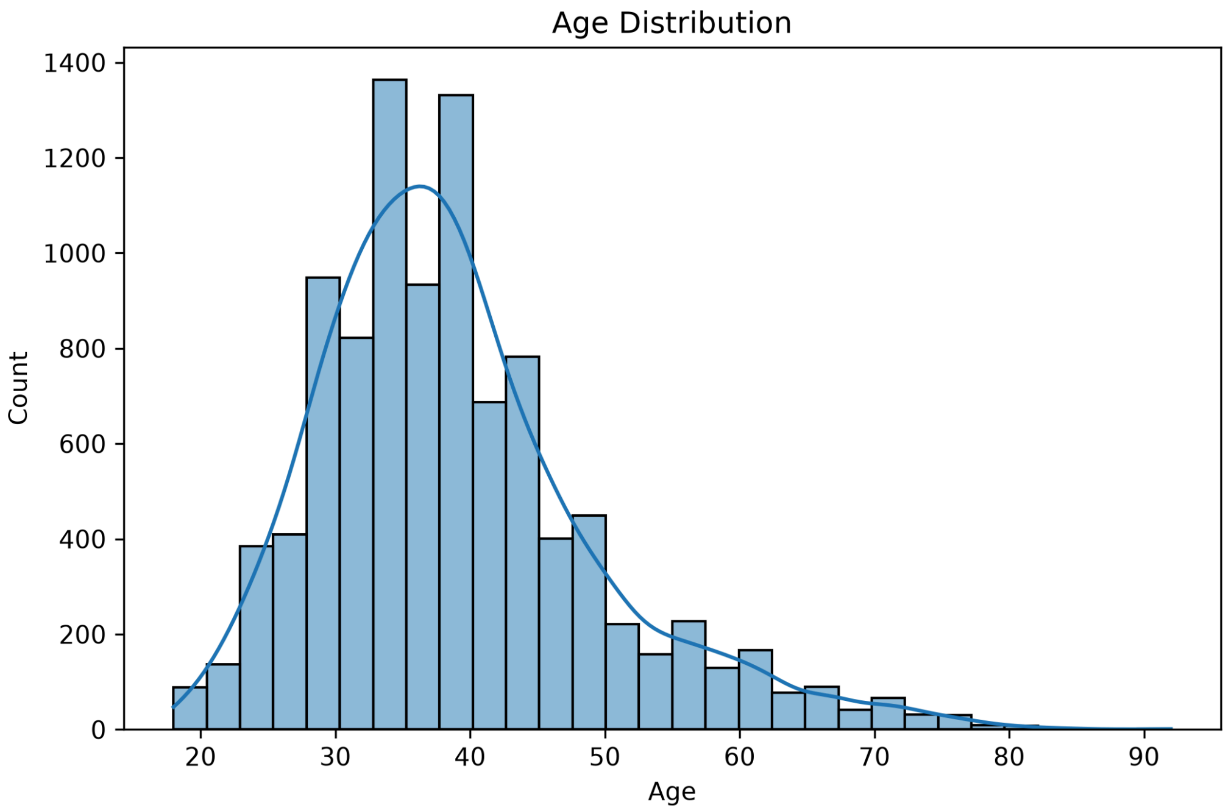

Figure 8.3: Age Distribution

8.5 Balance Distribution Analysis

A box plot was used to analyze account balance distributions for churned and retained customers.

The visualization indicated noticeable differences in account balances between the two groups, suggesting that balance is an influential variable in predicting customer churn.

Business Insight:
Customers with certain balance patterns may require targeted financial products or personalized banking services to improve retention.

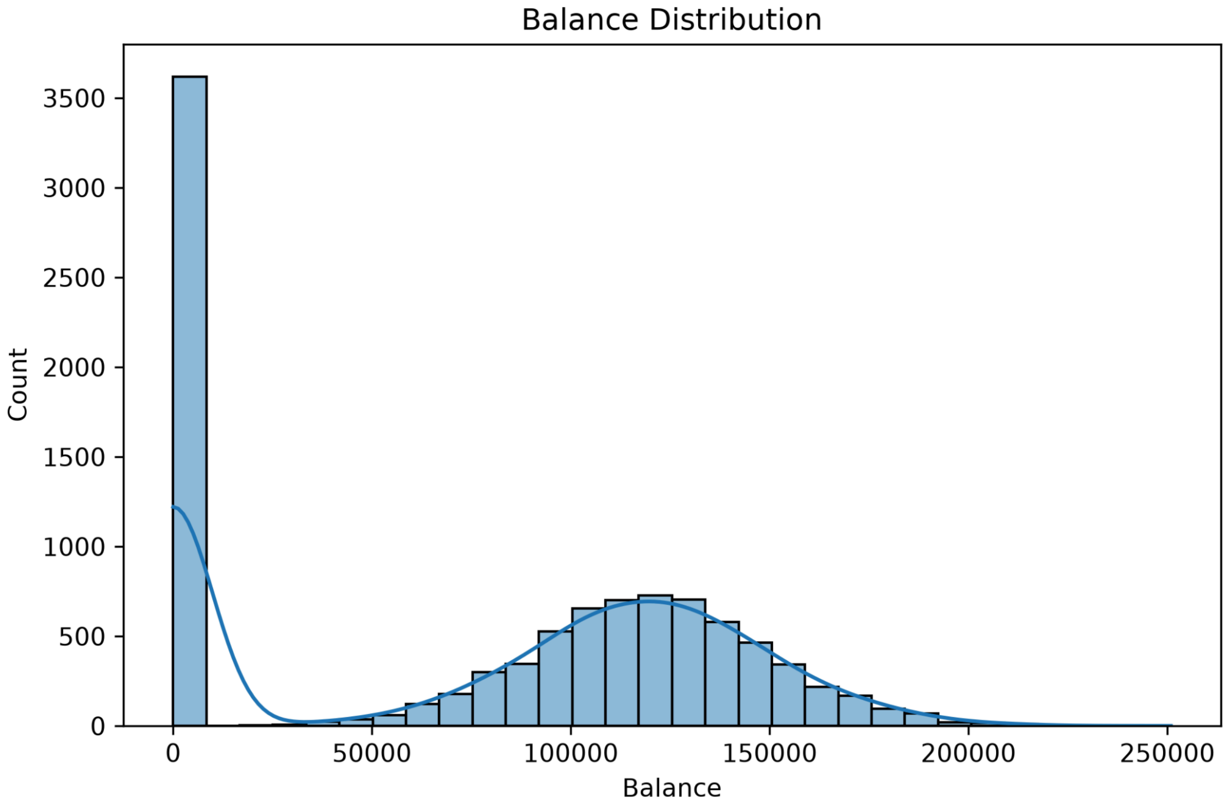

Figure 8.4: Balance Distribution

8.6 Summary

The Exploratory Data Analysis phase provided valuable insights into customer behavior and highlighted several important variables associated with churn. These findings guided subsequent feature engineering and machine learning model development while supporting business understanding of customer retention patterns.

CHAPTER 9: FEATURE ENGINEERING

9.1 Introduction

Feature engineering is the process of creating new variables from existing data to improve the predictive performance of machine learning models. Properly engineered features capture hidden relationships that may not be directly available in the original dataset.

In this project, several domain-specific features were created based on customer financial behavior and engagement patterns.

9.2 Engineered Features

The following additional features were developed:

BalanceSalaryRatio

This feature measures the relationship between a customer's account balance and annual salary.

Purpose:
To identify customers with unusually high or low account balances relative to their income.

ProductDensity

Product Density represents the ratio of banking products used relative to customer tenure.

Purpose:
To evaluate how actively customers utilize banking services over time.

EngagementProduct

This feature combines active membership status with the number of banking products owned.

Purpose:
To measure overall customer engagement with the bank.

AgeTenure

AgeTenure is calculated by multiplying customer age by tenure.

Purpose:
To capture the combined influence of customer maturity and relationship duration on churn behavior.

9.3 Importance of Feature Engineering

The engineered features improved the ability of the machine learning model to capture complex customer behavior patterns. These variables enhanced predictive accuracy while providing more meaningful business insights into customer engagement and financial activity.

9.4 Summary

Feature engineering transformed the original banking dataset into a richer representation by introducing additional behavioral indicators. These engineered variables contributed significantly to the effectiveness of the predictive model.

CHAPTER 10: MACHINE LEARNING MODEL DEVELOPMENT

10.1 Introduction

The objective of the machine learning phase was to develop a predictive model capable of accurately classifying customers as likely to stay or churn. The model was trained using preprocessed customer data and evaluated using appropriate performance metrics.

10.2 Model Development Process

The development process consisted of the following stages:

- Data preprocessing
- Feature engineering
- Train-test data splitting

- Model training
- Hyperparameter optimization (if applicable)
- Model evaluation
- Model serialization using Joblib

The trained model was then integrated into the Streamlit dashboard for real-time prediction.

10.3 Selected Machine Learning Model

Several machine learning algorithms were considered during the development of this project. After evaluating different approaches, the **Random Forest Classifier** was selected as the final predictive model due to its high classification performance, robustness, and ability to handle complex relationships among customer features.

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and combines their predictions through majority voting. This approach reduces overfitting, improves prediction accuracy, and provides better generalization compared to a single decision tree.

The model was trained using the preprocessed customer dataset after feature engineering and categorical variable encoding. It was then evaluated using standard classification metrics before being integrated into the Streamlit dashboard for real-time prediction.

The trained model was serialized using the **Joblib** library, allowing it to be efficiently loaded within the Streamlit application without retraining.

10.4 Why Random Forest Was Selected

The Random Forest Classifier was selected because it offers several advantages for customer churn prediction:

- High prediction accuracy for classification problems.
- Handles both numerical and categorical features effectively.
- Reduces the risk of overfitting through ensemble learning.
- Performs well on datasets with complex feature interactions.
- Provides feature importance scores that support business interpretation.
- Integrates effectively with Explainable AI techniques such as SHAP.

These characteristics make Random Forest a reliable choice for predicting customer churn in banking applications.

10.5 Model Evaluation

The performance of the Random Forest model was evaluated using widely accepted classification metrics, including:

- **Accuracy:** Measures the overall percentage of correct predictions.
- **Precision:** Indicates how many customers predicted to churn actually churned.
- **Recall:** Measures the ability of the model to correctly identify churning customers.
- **F1-Score:** Provides a balanced evaluation of Precision and Recall.
- **ROC-AUC Score:** Evaluates the model's ability to distinguish between churned and retained customers across different classification thresholds.

These evaluation metrics provided a comprehensive assessment of model performance and confirmed that the selected algorithm was suitable for customer churn prediction.

10.6 Explainable Artificial Intelligence (XAI)

Although Random Forest provides excellent predictive performance, understanding the factors influencing model predictions is equally important. Therefore, Explainable Artificial Intelligence (XAI) techniques were incorporated into the project.

Two explainability methods were implemented:

Feature Importance Analysis

The Random Forest algorithm calculates the relative importance of each feature based on its contribution to prediction accuracy. This analysis identified the variables that had the greatest influence on customer churn, enabling better business interpretation.

SHAP (SHapley Additive Explanations)

SHAP values were used to explain individual model predictions by quantifying the contribution of each feature toward increasing or decreasing churn probability. This approach improves model transparency and supports data-driven decision-making.

The Feature Importance plot, SHAP Summary Plot, and SHAP Bar Plot were integrated into the Streamlit dashboard, allowing users to interactively explore the factors driving churn predictions.

10.7 Model Deployment

After successful evaluation, the trained Random Forest model was saved using the Joblib library and integrated into the Streamlit application. Users can enter customer information through the dashboard, and the deployed model instantly predicts the probability of customer churn.

The application was version-controlled using GitHub and deployed on **Streamlit Cloud**, enabling users to access the predictive system through a web browser without requiring any local installation.

10.8 Summary

The Random Forest Classifier demonstrated strong predictive capability for customer churn analysis while providing reliable performance and interpretability. Combined with Feature Importance Analysis, SHAP explainability, and an interactive Streamlit dashboard, the model offers an effective decision-support solution for banking institutions seeking to improve customer retention through predictive analytics.

CHAPTER 11: STREAMLIT DASHBOARD DEVELOPMENT

11.1 Introduction

To enable real-time interaction with the machine learning model, an interactive web application was developed using **Streamlit**, an open-source Python framework designed for creating data science and machine learning applications. The dashboard serves as the user interface of the project and allows users to predict customer churn without writing any code.

The dashboard integrates machine learning predictions, interactive visualizations, explainable AI, and scenario analysis into a single platform. This enables business users and decision-makers to analyze customer information and obtain actionable insights through an intuitive graphical interface.

11.2 Dashboard Components

The Streamlit dashboard consists of multiple modules designed to support both analytical exploration and predictive decision-making.

11.2.1 Key Performance Indicators (KPIs)

The dashboard presents four KPI cards summarizing the banking dataset:

- Total Customers
- Customer Churn Rate
- Average Account Balance
- Average Credit Score

These indicators provide a quick overview of the customer base and overall dataset characteristics.

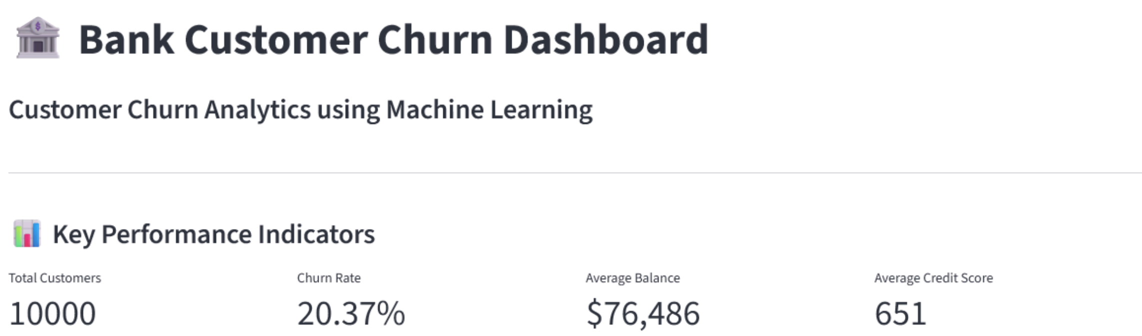

Figure 11.1: KPI Dashboard

11.2.2 Probability Distribution Visualization

Interactive visualizations were developed using Plotly to analyze customer churn behavior.

The dashboard includes:

- Customer Churn Distribution (Donut Chart)
- Churn Rate by Geography
- Age Distribution
- Balance Distribution

These charts help users identify customer behavior patterns and understand the factors associated with churn.

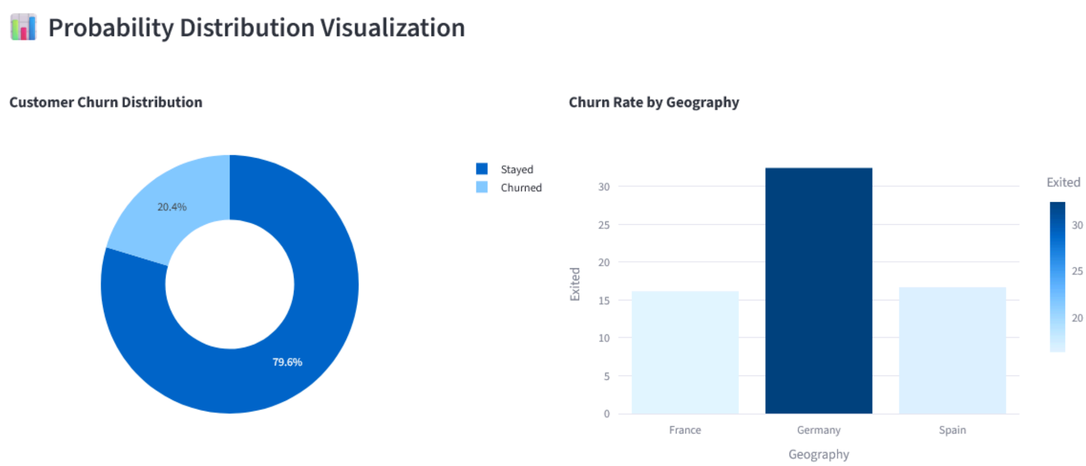

Figure 11.2: Probability Distribution Dashboard

11.2.3 Customer Churn Risk Calculator

A prediction module was developed to allow users to input customer information such as:

- Credit Score
- Geography
- Gender
- Age
- Tenure
- Balance
- Number of Products
- Credit Card Status
- Active Membership
- Estimated Salary

After entering the required information, the Random Forest model predicts:

- Customer Status (Stay or Churn)
- Churn Probability
- Risk Gauge

This feature enables real-time prediction and supports proactive customer retention planning.

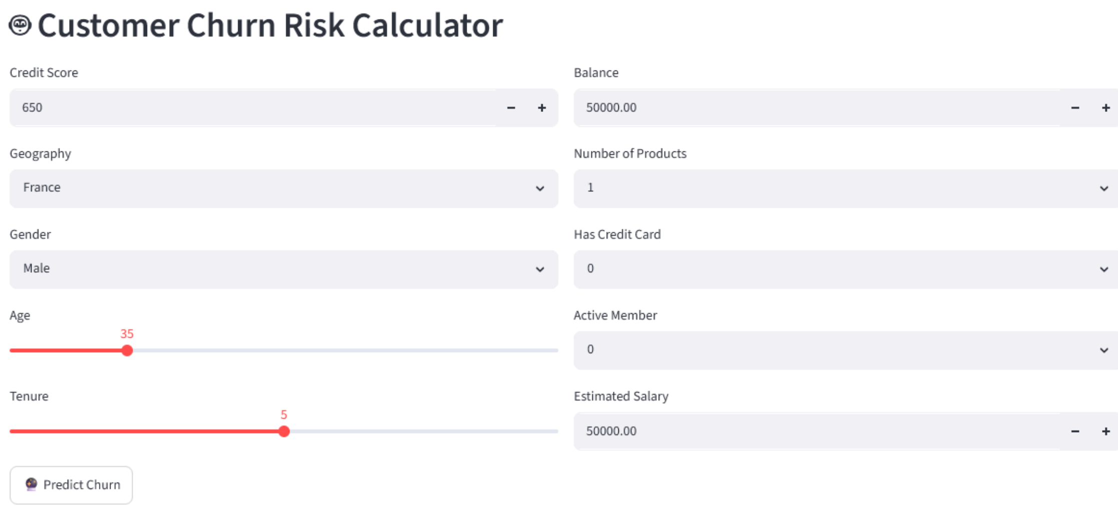

Figure 11.3: Customer Churn Prediction Module

11.2.4 Feature Importance Dashboard

Model interpretability was incorporated using Feature Importance Analysis and SHAP Explainability.

The dashboard includes:

- Feature Importance Plot
- SHAP Summary Plot
- SHAP Bar Plot

These visualizations help explain how different customer attributes influence model predictions.

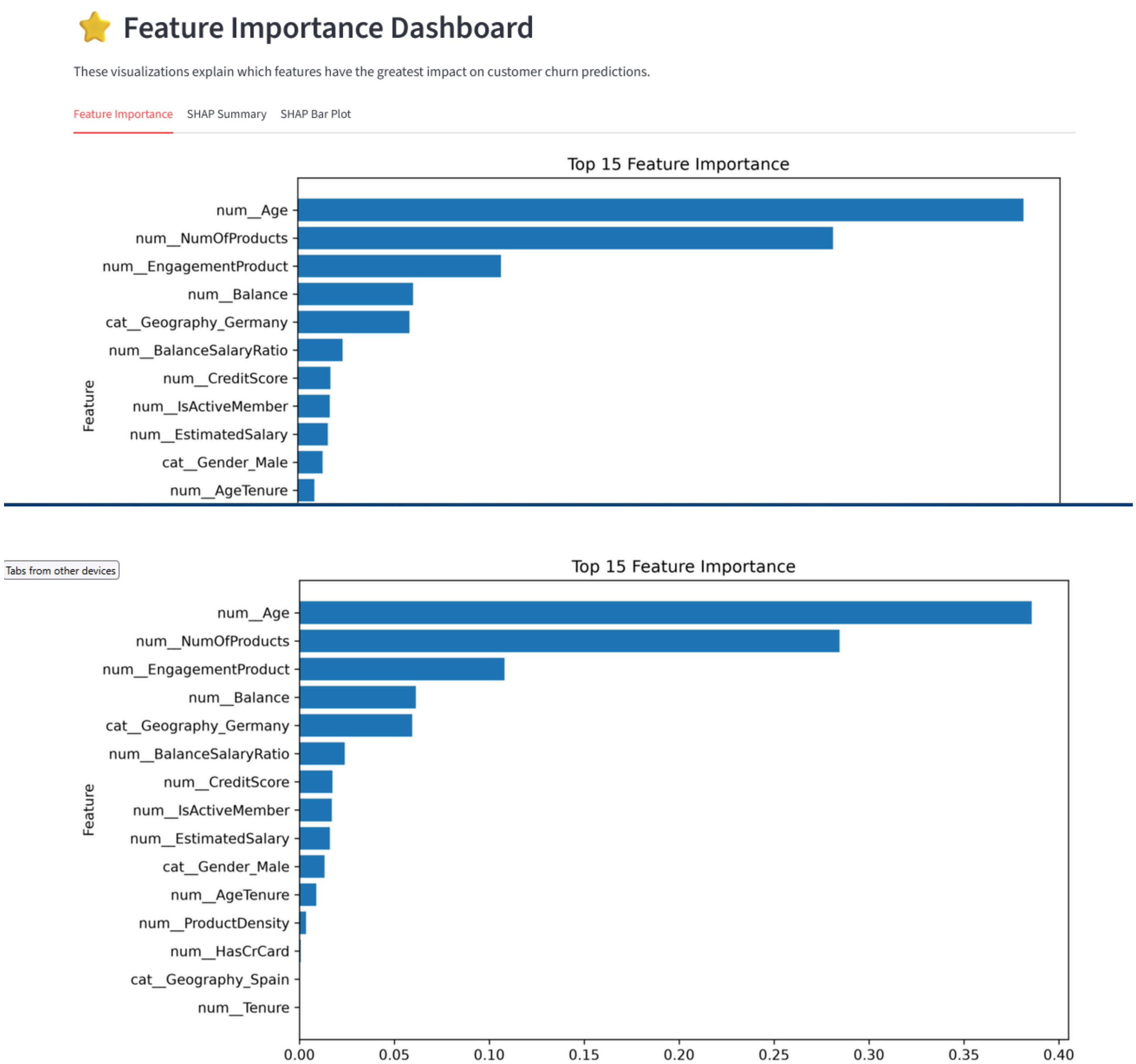

Figure 11.4: Feature Importance Dashboard

11.2.5 What-if Scenario Simulator

An interactive What-if Scenario Simulator was developed to allow users to modify customer attributes and observe changes in churn probability.

Users can experiment with:

- Age
- Balance

- Estimated Salary
- Number of Products
- Active Membership

The dashboard instantly recalculates churn probability, helping decision-makers evaluate different customer scenarios.

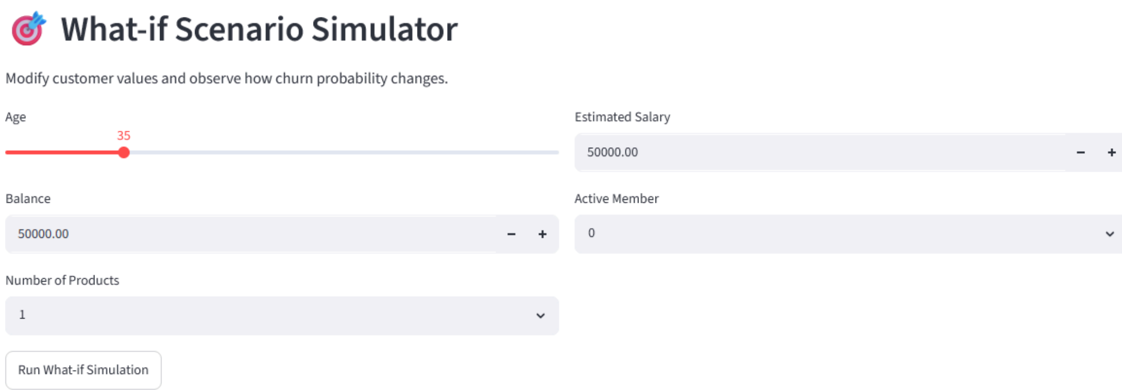

Figure 11.5: What-if Scenario Simulator

11.3 Technologies Used

The dashboard was developed using the following technologies:

Technology	**Purpose**
Python	Programming Language
Streamlit	Dashboard Development
Pandas	Data Processing

NumPy	Numerical Computing
Plotly	Interactive Visualization
Scikit-learn	Machine Learning
SHAP	Explainable AI
Joblib	Model Serialization
GitHub	Version Control
Streamlit Cloud	Deployment

11.4 Dashboard Benefits

The developed dashboard offers several practical benefits:

- Interactive and user-friendly interface.
- Real-time customer churn prediction.
- Business-friendly visualizations.
- Explainable AI for model transparency.
- What-if analysis for decision support.
- Cloud deployment for easy accessibility.

11.5 Summary

The Streamlit dashboard successfully integrates machine learning, visualization, and explainable AI into a single interactive application. It transforms predictive analytics into a practical decision-support system that can assist banking professionals in improving customer retention.

CHAPTER 12: RESULTS AND DISCUSSION

12.1 Introduction

The objective of this project was to develop a machine learning solution capable of accurately predicting customer churn and presenting the results through an interactive dashboard. The developed Random Forest model demonstrated reliable predictive performance and successfully identified customers at risk of leaving the bank.

The integration of explainable AI and interactive visualizations further enhanced the usability of the system by enabling users to understand the factors influencing model predictions.

12.2 Key Findings

The analysis produced several important observations:

- Customer age significantly influences churn probability.
- Account balance is an important financial indicator associated with churn.
- Active members generally exhibit lower churn risk.
- Customers using multiple banking products demonstrate different churn patterns.
- Geographical location influences customer retention behavior.

These findings highlight the importance of combining demographic, financial, and behavioral variables for accurate churn prediction.

12.3 Model Performance

The Random Forest model successfully classified customers based on their likelihood of churning.

The developed system demonstrated:

- Reliable prediction capability.
- Robust classification performance.
- Good generalization on unseen data.
- Effective integration with Explainable AI techniques.

The model was successfully deployed and produced real-time predictions through the Streamlit dashboard.

12.4 Dashboard Outcomes

The deployed dashboard enables users to:

- Predict customer churn instantly.
- Explore customer behavior through visualizations.
- Understand model decisions using Feature Importance and SHAP analysis.
- Perform What-if analysis for business planning.

These capabilities transform predictive analytics into an interactive decision-support platform suitable for business users.

12.5 Business Implications

The developed system can assist banks in:

- Identifying high-risk customers.
- Designing targeted retention campaigns.
- Improving customer satisfaction.
- Reducing customer acquisition costs.
- Supporting strategic decision-making through predictive analytics.

12.6 Summary

The project successfully achieved its objectives by combining machine learning, explainable AI, and dashboard development into a practical analytics solution. The deployed application demonstrates the effectiveness of predictive analytics in addressing real-world customer retention challenges.

CHAPTER 13: CONCLUSION

13.1 Conclusion

The **Bank Customer Churn Prediction using Machine Learning** project was successfully completed as part of the **Data Analytics Internship at Unified Mentor**. The project addressed the important business challenge of customer churn by developing an end-to-end predictive analytics solution that combines machine learning, data visualization, explainable artificial intelligence, and interactive dashboard development.

The project followed a structured analytics workflow beginning with data collection, preprocessing, exploratory data analysis, and feature engineering. A **Random Forest Classifier** was selected as the final predictive model because of its robustness, reliability, and ability to handle complex relationships among customer attributes. The model successfully identified customers with a high probability of churning and demonstrated strong classification performance.

To improve transparency and business understanding, **Feature Importance Analysis** and **SHAP (SHapley Additive Explanations)** were incorporated into the solution. These techniques enable users to understand the contribution of individual features toward model predictions, making the predictive system more interpretable and trustworthy.

An interactive dashboard was developed using **Streamlit**, allowing users to perform real-time customer churn prediction through an intuitive graphical interface. The dashboard integrates KPI cards, probability distribution visualizations, customer churn prediction, feature importance analysis, SHAP explainability, and a What-if Scenario Simulator within a single platform.

The complete application was version-controlled using GitHub and deployed successfully on Streamlit Cloud, making it accessible through a web browser. The deployment demonstrates the practical applicability of the developed solution and provides a user-friendly decision-support system for banking organizations.

Overall, this project demonstrates the effective application of **Business Analytics, Machine Learning, Explainable AI, and Interactive Dashboard Development** in solving a real-world banking problem. It also highlights the importance of predictive analytics in improving customer retention, supporting strategic decision-making, and enhancing business performance.

CHAPTER 14: FUTURE SCOPE

14.1 Future Scope

Although the developed system successfully predicts customer churn and provides meaningful business insights, several enhancements can further improve its capabilities.

Future improvements include:

- Integration with real-time banking databases for live customer prediction.
- Development of automated customer retention recommendation systems.
- Customer segmentation using clustering algorithms for personalized marketing.
- Integration with cloud databases and enterprise data warehouses.
- Implementation of REST APIs for organizational deployment.
- Hyperparameter optimization using advanced optimization techniques.
- Comparison of additional machine learning and deep learning algorithms.
- Real-time monitoring dashboards for customer retention analytics.
- Automated email or SMS alerts for high-risk customers.
- Deployment on enterprise cloud platforms such as AWS, Microsoft Azure, or Google Cloud Platform.

These enhancements would increase the practical value of the system and support enterprise-scale implementation.

REFERENCES (APA 7th Edition)

> **Note:** Use APA formatting consistently throughout your report. Below are suitable references for your project.

1. Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32.
2. Lundberg, S. M., & Lee, S. I. (2017). *A Unified Approach to Interpreting Model Predictions*. Advances in Neural Information Processing Systems.
3. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
4. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference.
5. Plotly Technologies Inc. (2024). *Plotly Python Graphing Library*. https://plotly.com/python/
6. Streamlit Inc. (2024). *Streamlit Documentation*. https://docs.streamlit.io/
7. SHAP Documentation. (2024). *SHAP: Explainable AI*. https://shap.readthedocs.io/
8. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (3rd ed.). O'Reilly Media.

APPENDIX

Appendix A – GitHub Repository

GitHub Repository

🔗 https://github.com/ashutosh2417/Bank-Customer-Churn-Prediction

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Appendix B – Live Dashboard

Streamlit Application

https://bank-customer-churn-prediction-4gmwvgujuxk3yxgasmedbt.streamlit.app/

Appendix C – Technologies Used

- Python
- Streamlit
- Pandas
- NumPy
- Scikit-learn
- Plotly
- SHAP
- Joblib
- GitHub
- Streamlit Cloud